Pratik Nanaso Raut

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PROFESSIONAL SUMMARY

MCA candidate with strong foundation in Java, Spring Boot, and REST APIs — adept at developing scalable backend applications.

Hands-on experience building multi-tenant payroll systems and student management portals using Java, Spring Boot, MySQL, and MongoDB in Agile development environments with focus on Database Optimization and Object-Oriented Programming.

TECHNICAL SKILLS

Languages	Java, SQL, JavaScript, C, C++, Python (basics)
Backend	Spring Boot, REST APIs, Spring Security, Hibernate, JDBC, JWT
Databases	MySQL, MongoDB
Tools	Git, GitHub, Maven, Postman, IntelliJ IDEA
Concepts	Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), SDLC, REST API Design, Database Optimization, RBAC, JWT Authentication, Multithreading
Exposure	Cloud computing basics (AWS), AI/ML fundamentals, Big Data concepts

PROFESSIONAL EXPERIENCE

Software Developer Intern *Feb 2026 -- Ongoing*

SetTribe IT Solutions | Pune

- Developed ERP application modules using Java and Spring Boot, implementing RESTful API contracts for scalable service architecture.
- Operated in Agile sprint workflow using Git for version control, code reviews, sprint planning, and technical documentation.
- Designed and tested application modules with systematic debugging, reducing defect reopen rate by 40%.

ACADEMIC PROJECTS

PayStride -- Multi-Tenant SaaS Payroll Management System *Java, Spring Boot, MySQL, Hibernate, REST API, JWT*

- Architected multi-tenant RBAC system using Spring Security and JWT, enforcing complete data isolation across all API endpoints.
- Optimized MySQL query execution through strategic indexing and schema normalization, reducing average response latency by 40%.
- Engineered BigDecimal payroll computation engine handling statutory deductions (PF 12%, ESI 0.75%) and 1.5x overtime multipliers with zero precision loss.

EduTrack -- Student Management Portal *Java, Spring Boot, MongoDB, REST API*

- Developed RESTful CRUD APIs over Spring Boot handling 500+ student records with optimized MongoDB aggregation queries.
- Designed modular service architecture enabling independent scaling of student, batch, and reporting modules.
- Implemented dual-layer validation (client-side and server-side) reducing data entry errors by 25%.

Customised Virtual File System (CVFS) -- In-Memory Linux-like File Manager *C, C++, Data Structures, OS Concepts*

- Simulated Linux file system operations (open, read, write, lseek) in C++ using inode-based metadata management and a custom shell interpreter.
- Achieved O(1) file lookup by designing hash table-based inode indexing with chaining for collision resolution.
- Implemented in-memory file allocation table using C++ classes managing file permissions, ownership, and hierarchical directory traversal.

TECHNICAL STRENGTHS

- Proficient in backend application development, REST API design, and database-driven systems.
- Hands-on experience building concurrent, multi-tenant, and performance-optimized Java applications.
- Experience in Agile development workflows, Git-based version control, and collaborative engineering.
- Strong debugging, query optimization, and software architecture capabilities.

EDUCATION

Master of Computer Applications (MCA) *2026 | Pursuing*

Dnyansagar Institute of Management and Research, Pune (SPPU) 8.25 CGPA | Sem VII: 7.24 SGPA

Bachelor of Computer Applications (BCA) *2024*

C.T. Bora College, Shirur (SPPU) 73.34%

Higher Secondary Certificate (HSC) *2021*

Maharashtra State Board 75%